# OmniClass: Next-Generation LMS

A Full-Stack AI-Powered Platform for Digital Learning & Institutional Governance

| Attribute | Details |
|---|---|
| **Domain** | Full Stack Web Development (MERN + Applied AI) |
| **Project Type** | Individual Major Project |
| **Duration** | 10 Days |
| **Difficulty Level** | Advanced / Intermediate |
| **Primary Focus** | Frontend + Backend + Database + Role-Based Authentication + Multi-Agent AI + Interactive Dashboards + Cryptographic Verification |
| **Platform Name** | **OmniClass** |
| **Repository** | https://github.com/Shaik673/OmniClass.git |

## PROJECT GOAL

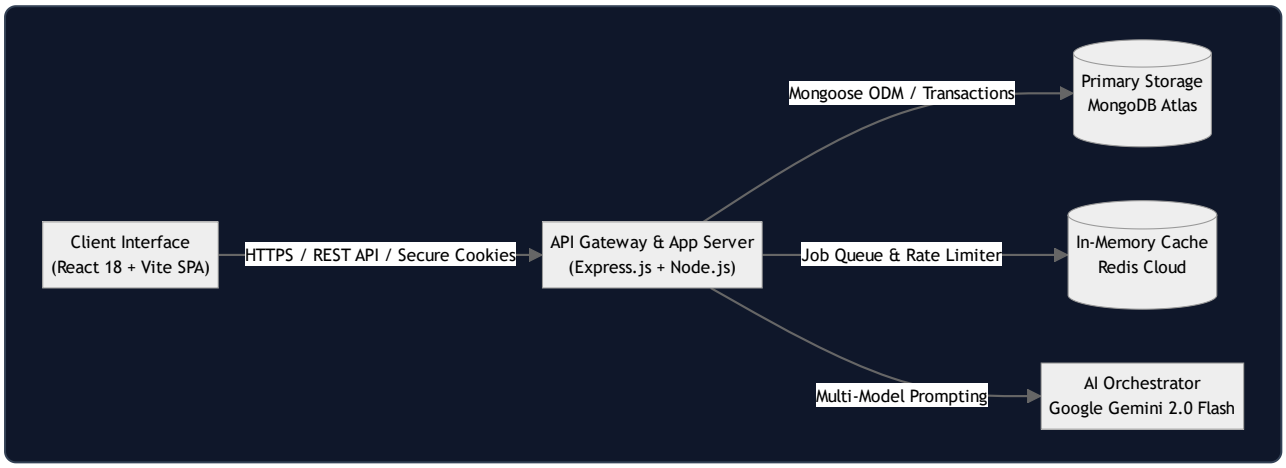
Design, architect, and deploy an enterprise-grade, full-stack Learning Management System (**OmniClass**) where learners can register, browse, and enroll in curated academic courses, navigate interactive learning modules, submit assignments for formative evaluation, engage with an AI study tutor, take 30-MCQ proctored exams, and earn cryptographically validated A4 landscape digital certificates.

Simultaneously, provide administrators and instructors with comprehensive dashboards to author multi-module curricula, manage enrollments, evaluate submissions, track student mastery analytics, and oversee institutional governance.

## SECTION 1 Project Overview

The **OmniClass Learning Management System** is a unified full-stack web application engineered to bridge the divide between traditional asynchronous coursework and real-time intelligent tutoring. The platform centralizes educational delivery, formative assessments, student engagement, and credentialing into a cohesive, high-performance ecosystem.

OmniClass connects three distinct stakeholders—**Students**, **Instructors**, and **Institutional Administrators**—through granular, role-based access controls and customized interfaces.




### Core Full-Stack Flow

1. **Frontend Presentation Layer**: Built on React 18 with Vite, Tailwind CSS, and Framer Motion. Provides accessible light/dark themes, responsive mobile drawer navigation, letter-by-letter live AI text streaming, audio voice playback, and interactive charts.
2. **Backend Application Layer**: Built on Node.js and Express.js with a modular layered architecture ( `controllers` , `routes` , `middleware` , `models` , `services` , `jobs` ). Enforces rate limiting, helmet security headers, request tracing, and input validation.
3. **Database & Persistence Layer**: Powered by MongoDB Atlas (NoSQL Document Store) for high-flexibility schema modeling with Mongoose ODM, accompanied by Redis Cloud for BullMQ scheduled jobs, task queues, and rate-limiting counters.
4. **AI Orchestration Layer**: Powered by Google Gemini 2.0 Flash API with multi-model fallback routines for real-time tutoring, YouTube/document summarization, 3-part active recall quiz synthesis, and contextual code generation.

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## SECTION 2 Project Objectives

1. **Unified Academic Ecosystem**: Provide a single centralized web portal for catalog browsing, module consumption, assessment submission, and grading.
2. **Robust Identity & Access Governance**: Implement stateful cryptographic JWT session authentication via `httpOnly` , `sameSite: none` , `secure` cookies with bcrypt password hashing (12 salt rounds) and OAuth2 social login (Google, GitHub, Facebook).
3. **Curriculum & Content Management**: Allow instructors and administrators to create, edit, reorder, and publish courses with structured 5-module sequences, markdown notes, PDF materials, code snippets, and video links.

4. **Formative & Proctored Assessments**: Provide automated grading for 3-part AI quizzes and 30-MCQ proctored final exams with strict 70% passing criteria required to unlock completion credentials.
5. **Cryptographic Certificate Engine**: Automatically issue tamper-evident A4 landscape certificates with unique serial numbers, institutional seals, instructor signatures, and scannable QR verification codes linking to public verification routes.
6. **Interactive AI Study Room**: Integrate Omni Tutor with letter-by-letter typewriter streaming, expressive voice synthesis that strips special symbols, and YouLearn-style 3-mode content summarization (File Drop, YouTube URL, Manual Paste).
7. **Production Engineering & Reliability**: Demonstrate end-to-end full-stack development, comprehensive testing suites, automated seeding, and free production deployment across Vercel (Edge CDN) and Render (Web Service).

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## SECTION 3 User Roles & RBAC Matrix

| Capability / Permission | Student | Instructor | Administrator |
|---|---|---|---|
| Self-Registration & Email Verification | Yes | No (Provisioned) | No (Provisioned) |
| Course Catalog Browsing & Search | Yes | Yes | Yes |
| Course Enrollment (Free & Pro Tier) | Yes | Yes | Yes |
| Module Reading & Progress Tracking | Yes | Yes | Yes |
| AI Omni Tutor & Voice Q&A Access | Yes | Yes | Yes |
| 3-Mode Content Summarizer & AI Quiz | Yes | Yes | Yes |
| Assignment Submission (Files/URLs/Text) | Yes | View Only | View Only |
| 30-MCQ Final Proctored Exam & Certificate | Yes | Audit | Audit |
| Course Creation, Editing & Publishing | No | Yes | Yes |
| Module & Learning Material Authoring | No | Yes | Yes |
| Assignment Creation & Rubric Definition | No | Yes | Yes |
| Submission Grading & Formative Feedback | No | Yes | Yes |
| Student Roster & Academic Analytics | No | Yes (Assigned) | Yes (All) |
| Course Approval & Moderation | No | No | Yes |
| Institutional Settings & User Management | No | No | Yes |

## SECTION 4 Minimum Completion Criteria

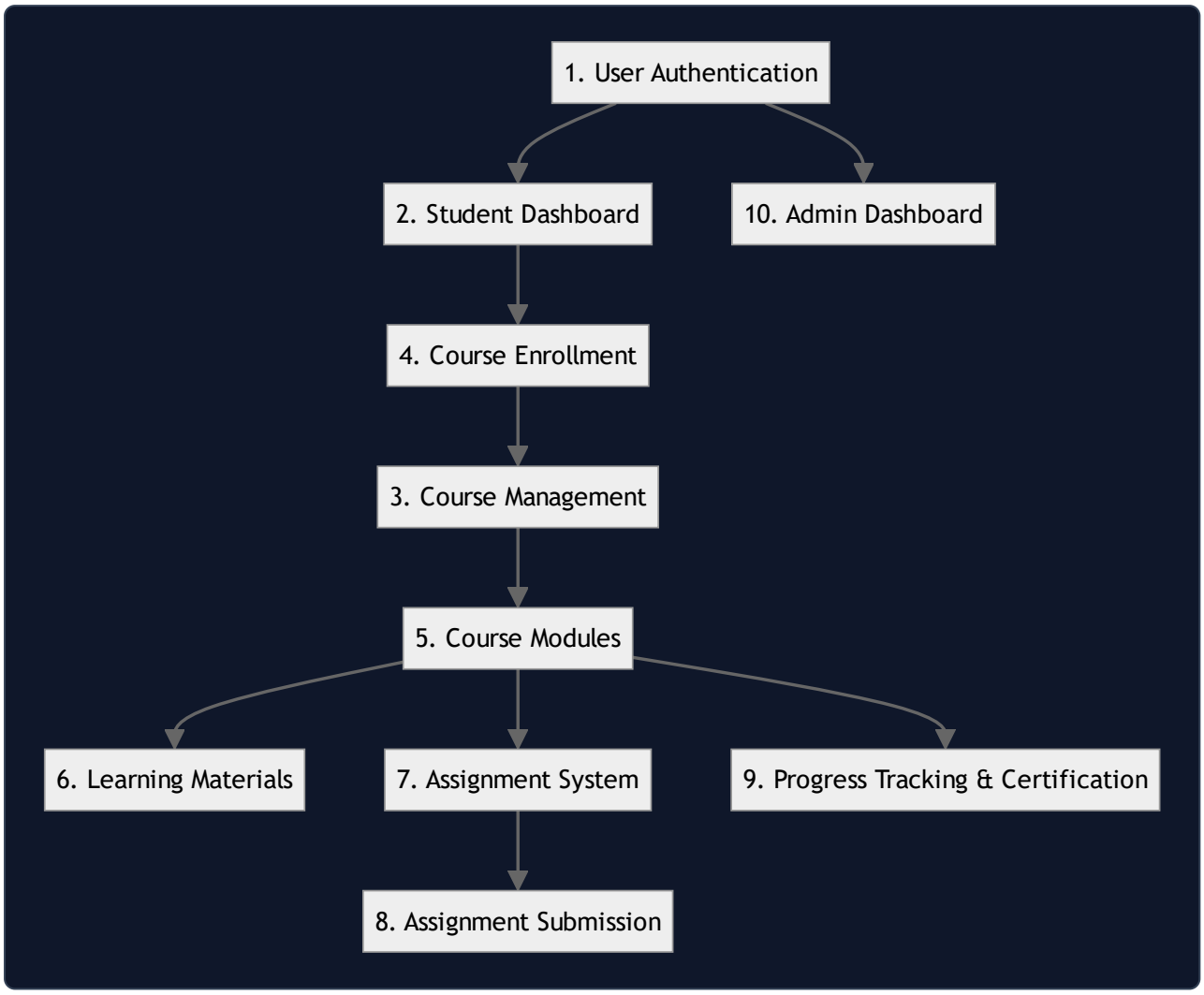
- ✓ **User Registration & Login**: Validated email/password authentication, JWT cookie persistence, and OAuth2 social integration.
- ✓ **Student/Admin/Instructor Roles**: Enforced at both route and database query levels.
- ✓ **Course Management**: Full CRUD operations for course entities with categories, duration, difficulty, and pricing.
- ✓ **Course Enrollment**: Tracking enrollment dates, completion status, exam attempts, and progress percentage.
- ✓ **Course Modules**: 5 structured modules per course with clean markdown, code cards, and downloadable references.
- ✓ **Assignment Management**: Assignment creation with deadlines, maximum scores, submission guidelines, and rubrics.
- ✓ **Assignment Submission**: Multi-mode submissions (GitHub repositories, live URLs, text answers, document attachments).
- ✓ **Progress Tracking**: Real-time progress percentage bar updated upon module and quiz completion.
- ✓ **Student Dashboard**: Enrolled courses, dynamic analytics charts, pending deadlines, and quick navigation.

- ✓ **Admin Dashboard**: System metrics, approval queues, user management, and institutional settings.
- ✓ **Database Integration**: MongoDB Atlas persistence with zero mock fallbacks in production.

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SECTION 5

# Functional Modules




## Module 1: User Authentication & Role Management

- Secure sign-up, login, password reset, and session termination.
- Passwords hashed with bcrypt (cost factor 12).
- Session issued via cryptographically signed JWT cookies ( `oc_session` ).
- Social authentication with Google, GitHub, and Facebook OAuth2.
- Session invalidation via `sessionVersion` counter upon password change.

## Module 2: Student Dashboard & Visual Analytics

- High-level overview displaying Enrolled Courses, Active Hours, Average Mastery Score, and Completed Modules.
- Recharts-powered graphical visualizations tracking weekly study velocity and subject distribution.
- Upcoming deadlines drawer for assignments and proctored evaluations.
- Quick-resume cards allowing one-click navigation to the learner's last accessed lesson.

### Module 3: Course Management Lifecycle

- Course authoring with title, slug, description, category, difficulty tier (Beginner, Intermediate, Advanced), duration, and thumbnail cover.
- Pro course tier with pricing (INR) and Razorpay order generation & signature verification.
- Approval workflow ensuring drafts are reviewed by Institutional Admins before publishing.

### Module 4: Course Enrollment & Subscription Engine

- Self-enrollment for free courses and payment verification for Pro courses.
- Enrollment state tracking: `active`, `completed`, `dropped`, or `pending`.
- Record keeping for module reading history, timestamps, exam scores, and certificate references.

### Module 5: Course Modules & Structured Curriculum

- Hierarchical course layout: `Course` -> `Modules` (minimum 5 per course) -> `Lessons`.
- Sequential unlocking logic preventing learners from skipping directly to final exams without reading module materials.
- Ordering controls allowing instructors to drag or adjust curriculum flow.

### Module 6: Learning Materials & Rich Markdown Engine

- Custom-styled Markdown reader supporting GitHub Flavored Markdown (tables, blockquotes, task lists).
- Syntax-highlighted code cards with language tags and one-click copy buttons.
- Supplementary resource attachments (PDF links, external documentation, references).

### Module 7: Assignment Creation & Management

- Instructor interface to publish assignments tied to specific courses.
- Assignment parameters: title, detailed brief, deadline picker, maximum points, and grading rubrics.
- State management: `draft`, `published`, `closed`, and `archived`.

### Module 8: Assignment Submission Engine

- Student submission portal accepting GitHub project links, Drive/Cloud URLs, and rich text commentary.
- Timestamp validation enforcing deadlines and flagging late submissions.
- Formative evaluation interface for instructors to record points, feedback commentary, and pass/resubmit status.

### Module 9: Progress Tracking & Final Certification

- Dynamic progress calculation: `(Completed Modules / Total Modules) * 100`.
- 30-MCQ Final Proctored Exam unlocked upon reaching 100% module reading progress.
- Server-side exam grading requiring ≥ 70% (21/30) to pass.
- Automated issuance of verified A4 landscape digital certificates with QR code verification and 1-click LinkedIn Certification sharing.

### Module 10: Admin Dashboard & Institutional Governance

- High-level KPI cards: Total Registered Learners, Active Faculty, Published Courses, Total Enrollments, System Health.
- Course approval queue for peer review.
- User management table with search, role modification, and account status toggling.

- Institutional profile branding (logo, domain, university motto, signatory title).

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## SECTION 6 Student Features

- **Account Security**: Create an account, verify email, and log in securely via password or OAuth.
- **Catalog Exploration**: Search, filter by category/difficulty, and view syllabus outlines and instructor bios.
- **Course Player**: Structured sidebar navigation through all 5 modules with completion tick marks and reading progress bar.
- **Interactive Omni Tutor**:
- Chat interface powered by Gemini 2.0 Flash.
- Letter-by-letter dynamic typewriter streaming.
- Multi-turn conversation memory with strict 5-session auto-pruning.
- Voice audio synthesis that strips dashes, asterisks, and code formatting to speak natural sentences.
- **Content Summarizer (YouLearn Style)**:
- Drop & Upload documents ( `.pdf` , `.docx` , `.ppt` , `.txt` ).
- Paste educational YouTube video URLs to generate timestamped key takeaways.
- Manual text paste with instant executive summary notes.
- **3-Part AI Quiz & Flashcard Generator**:
- Dynamic generation of 15 MCQs partitioned into **Part 1: Conceptual Foundations**, **Part 2: Applied Analysis**, and **Part 3: Advanced Problem Solving**.
- Interactive double-sided flip flashcards for active recall.
- **Assignment Hub**: View guidelines, submit repository links, and review instructor grades with formative feedback.
- **30-MCQ Final Exam**: Timed proctored assessment testing full course concepts.
- **A4 Landscape Certificate**:
- High-resolution modal with gold double borders, university seal, authorized signature, and unique certificate ID.
- Live QR code linking to `/verify/:certificateCode` .
- `@media print` landscape A4 stylesheet for high-quality PDF downloads.
- 1-click LinkedIn Certification dialog pre-filled with credentials.
- **Profile Customization**:
- Upload custom hero cover banners ( `.jpg` / `.png` ).
- LeetCode profile validation via GraphQL with verification challenge tokens ( `omniclass-verify-XXXXXX` ).

## SECTION 7 Admin & Instructor Features

- **Instructor Copilot**: AI-assisted syllabus generator, quiz authoring assistant, and assignment rubric drafter.
- **Course Builder**: Intuitive visual editor to write module content, embed code blocks, attach references, and set Pro pricing.
- **Submissions Grading**: Centralized queue to inspect student repositories, verify live demo URLs, input grades, and return constructive feedback.
- **Student Roster**: Filter students by enrollment status, track completion velocity, and identify students needing academic intervention.
- **Administrative Moderation**:
- Course Approval Queue: Review instructor course drafts before public catalog release.

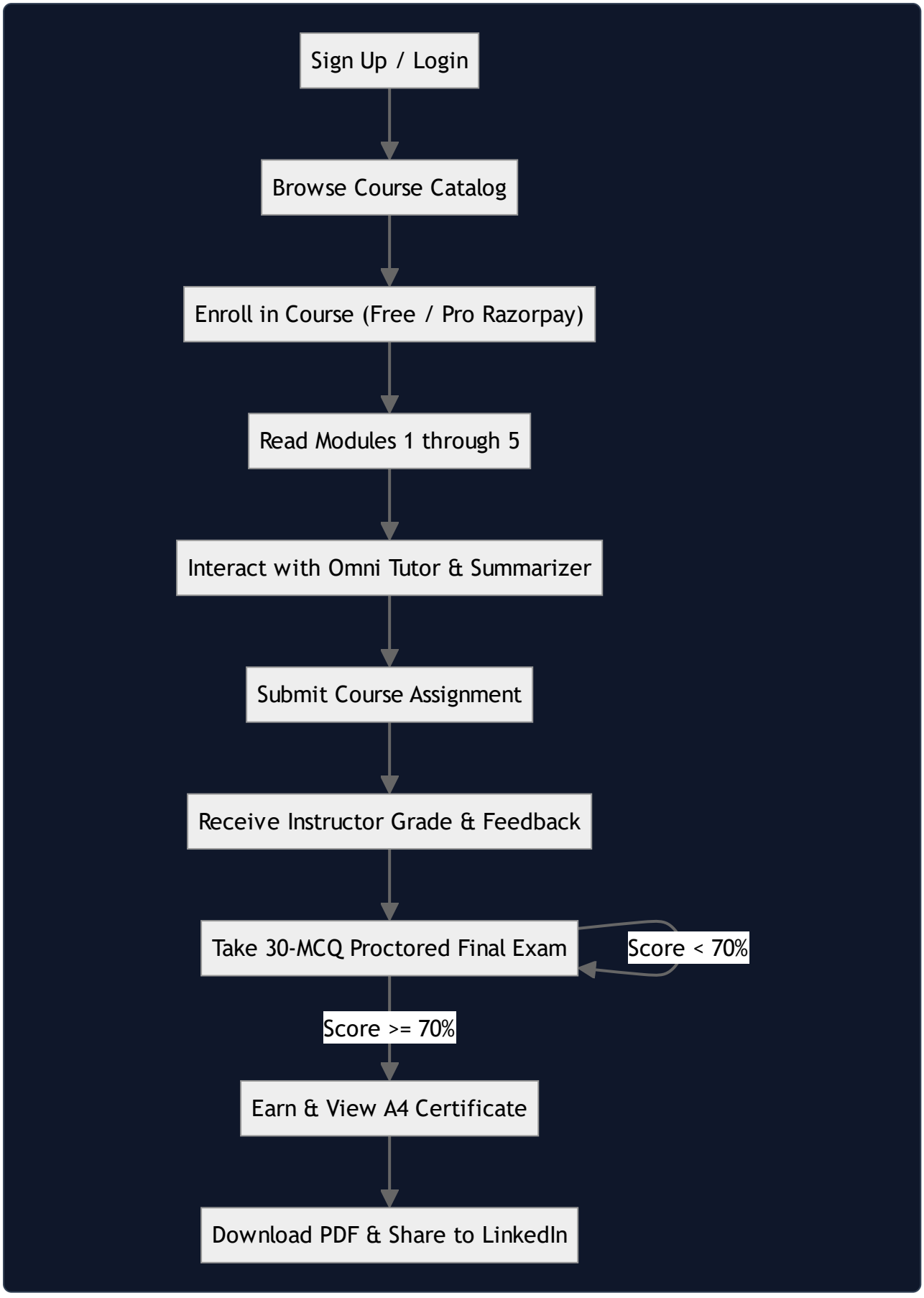
- Institutional Configuration: Update university name, branding colors, and accreditation disclosures.
- User Directory: Manage user roles ( `student` , `instructor` , `admin` ) and deactivate suspicious accounts.

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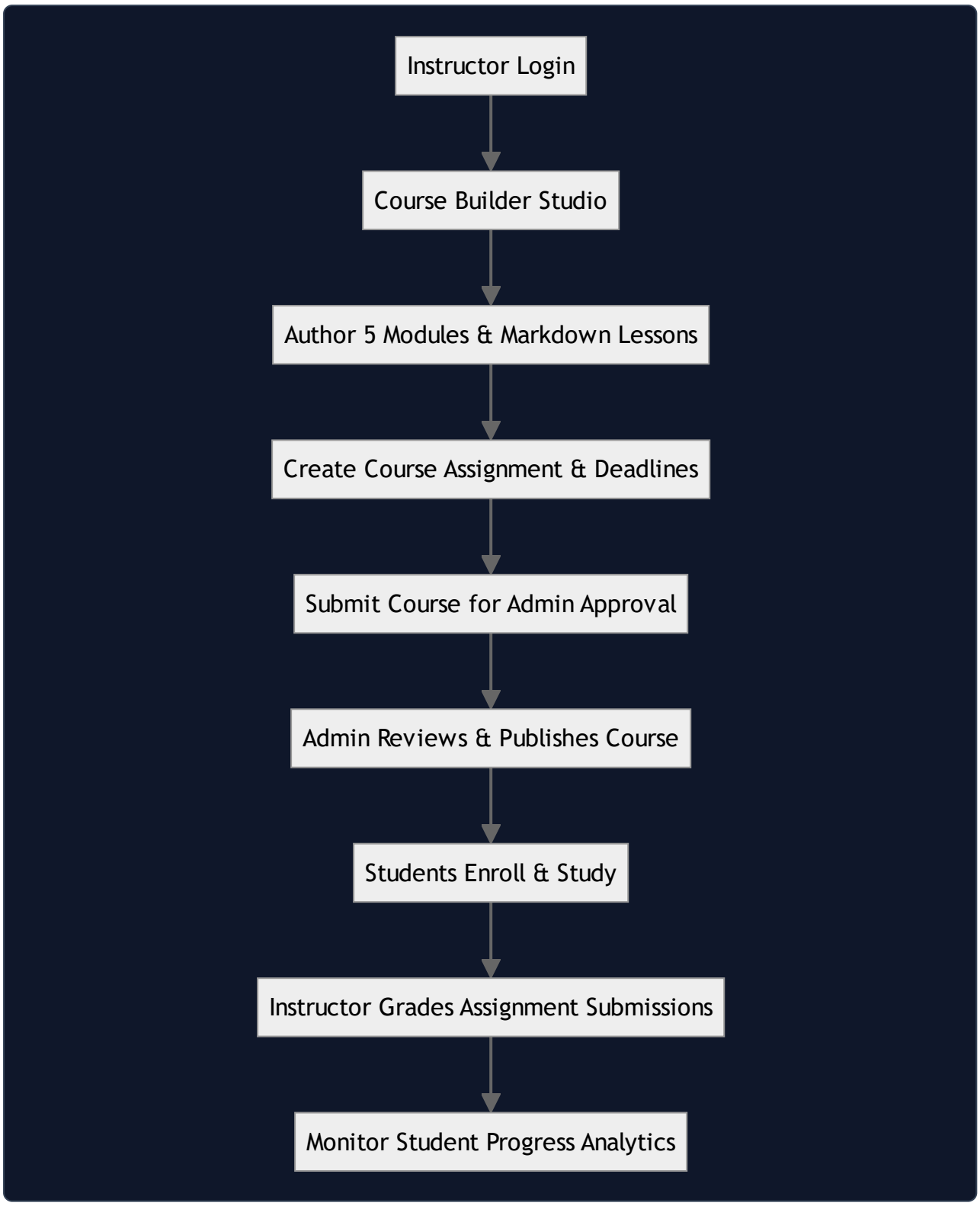
SECTION 8

# Expected User Workflows

## Student Workflow


Sign Up / Login
Browse Course Catalog
Enroll in Course (Free / Pro Razorpay)
Read Modules 1 through 5
Interact with Omni Tutor & Summarizer
Submit Course Assignment
Receive Instructor Grade & Feedback
Take 30-MCQ Proctored Final Exam
Score < 70%
Score >= 70%
Earn & View A4 Certificate
Download PDF & Share to LinkedIn

### Admin / Instructor Workflow



## SECTION 9 Implemented Production Courses

The platform is seeded with 4 complete courses containing 5 structured modules each:

```
├── Course 1: Algorithmic Problem Solving & Technical Interview Mastery (Free)
│   ├── Module 1: Foundations of Computational Complexity & Big-O Analysis
│   ├── Module 2: Non-Linear Structures: Trees, Tries & Priority Queues
│   ├── Module 3: Dynamic Programming: State Formulation & Memoization
│   ├── Module 4: Graph Theory: Traversals, Shortest Paths & Disjoint Sets
│   └── Module 5: System Design Essentials: Scalability, Caching & Sharding
│
├── Course 2: Cloud Systems Architecture & Distributed Computing (Pro &mdash; ₹1,999)
│   ├── Module 1: Distributed Systems Theory: CAP Theorem & Consensus
│   ├── Module 2: Containerization & Orchestration with Docker and Kubernetes
│   ├── Module 3: Event-Driven Architectures & Asynchronous Messaging
│   ├── Module 4: Serverless Computing & Cloud Infrastructure as Code
│   └── Module 5: Observability, SRE Principles & Zero-Trust Cloud Security
│
├── Course 3: Human-Centered UI/UX & Design Systems (Free)
│   ├── Module 1: Fundamentals of Cognitive Psychology & User Interaction
│   ├── Module 2: Wireframing, High-Fidelity Prototyping & User Testing
│   ├── Module 3: Design Systems Engineering: Tokens, Components & Patterns
│   ├── Module 4: Web Accessibility (WCAG 2.2 AA) & Inclusive Interfaces
│   └── Module 5: Micro-Interactions, Animation Performance & State Feedback
│
└── Course 4: Foundations of Artificial Intelligence & Machine Learning (Pro &mdash; ₹1,499)
    ├── Module 1: Statistical Learning, Mathematical Foundations & Optimization
    ├── Module 2: Supervised Learning: Regression, Classification & Ensembles
    ├── Module 3: Unsupervised Learning: Clustering & Dimensionality Reduction
    ├── Module 4: Deep Learning Foundations: Perceptrons, CNNs & Backprop
    └── Module 5: Large Language Models, Transformer Architecture & AI Ethics
```

SECTION 10

# Application Pages & Route Map

| Category | Page Name | Route Path | Description |
|---|---|---|---|
| **Public** | Landing Home | `/` | Hero section, accreditation badges, course promo, comprehensive 4-column footer, newsletter. |
| **Public** | Course Catalog | `/catalog` | Searchable grid of available courses with category filters and Pro price badges. |
| **Public** | Public Verification | `/verify/:code` | Cryptographic public ledger page verifying authenticity of issued certificates. |
| **Auth** | Sign In / Register | `/login`, `/register` | Secure authentication with social OAuth, email verification, and password validation. |
| **Student** | Student Dashboard | `/dashboard` | Enrolled courses, visual charts, study session tracking, and assignment alerts. |
| **Student** | Course Player | `/learn/:courseId` | 5-module reader, markdown code snippets, assignment submission, and 30-MCQ exam. |
| **Student** | AI Study Room | `/ai` | Omni Tutor, speech voice player, 3-mode Summarizer, and 3-part Quiz Generator. |
| **Student** | Certificates Page | `/certificates` | Repository of all earned credentials with A4 modal, PDF print, and LinkedIn share. |
| **Student** | User Profile | `/profile` | Hero banner upload, LeetCode verification challenge, and connected accounts. |
| **Instructor** | Teaching Dashboard | `/teaching` | Overview of instructor courses, active students, and pending grading queue. |
| **Instructor** | Course Builder | `/courses/create`, `/edit/:id` | Module management, markdown editor, resource attachment, and pricing toggle. |
| **Admin** | Admin Dashboard | `/admin` | System-wide metrics, institutional revenue, active users, and health monitors. |
| **Admin** | Course Approvals | `/admin/approvals` | Review workflow for newly authored courses before public catalog release. |
| **Admin** | User Management | `/admin/users` | User directory with role promotion, search, and status toggles. |

SECTION 11

# Technology Stack Selection

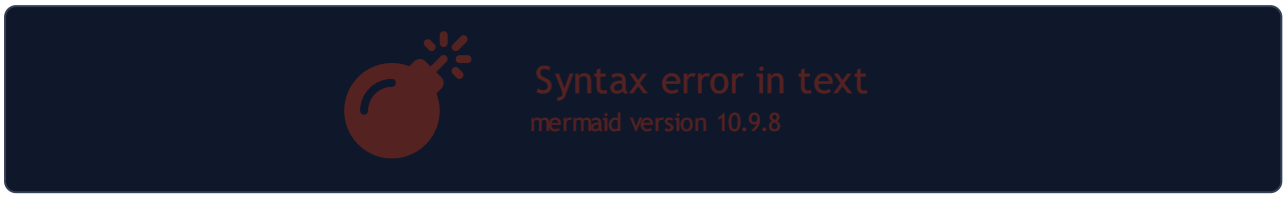



## Justification of Chosen Stack

- **React 18 + Vite**: High-speed Hot Module Replacement (HMR), tree-shaking, lightweight bundle sizes (under 320 kB gzipped), and reactive state management.
- **Node.js + Express.js**: Non-blocking asynchronous I/O ideal for concurrent tutoring streams, websocket notifications, and REST endpoints.
- **MongoDB Atlas**: Flexible document data model suited for complex hierarchical course content, variable quiz questions, and unstructured AI responses.
- **Redis Cloud**: Low-latency caching for session blacklists, rate-limiting tokens, and background queues via BullMQ.
- **Google Gemini 2.0 Flash**: State-of-the-art token throughput and 1,500 free daily requests, making intelligent tutoring scalable at zero cost.

SECTION 12

## System Architecture



SECTION 13

# Database Schema & Data Models

## 1. `User` Schema

```
{
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  passwordHash: { type: String, required: true },
  role: { type: String, enum: ['student', 'instructor', 'admin'], default: 'student' },
  institutionId: { type: ObjectId, ref: 'Institution' },
  avatar: { type: String },
  banner: { type: String },
  theme: { type: String, enum: ['dark', 'light', 'system'], default: 'dark' },
  status: { type: String, enum: ['active', 'suspended', 'pending'], default: 'active' },
  sessionVersion: { type: Number, default: 0 },
  integrations: {
    github: { connected: Boolean, username: String },
    linkedin: { connected: Boolean, profileUrl: String },
    leetcode: {
      connected: Boolean,
      username: String,
      verified: Boolean,
      contestRating: Number,
      stats: { totalSolved: Number, easy: Number, medium: Number, hard: Number, ranking: Number },
      verificationChallenge: String,
      challengeExpiresAt: Date
    }
  },
  createdAt: { type: Date, default: Date.now }
}
```

### 2. `Course` Schema

```
{
  title: { type: String, required: true },
  slug: { type: String, required: true, unique: true },
  description: { type: String, required: true },
  category: { type: String, required: true },
  difficulty: { type: String, enum: ['beginner', 'intermediate', 'advanced'], default: 'beginner' },
  instructor: { type: ObjectId, ref: 'User', required: true },
  institutionId: { type: ObjectId, ref: 'Institution', required: true },
  durationHours: { type: Number, default: 10 },
  thumbnail: { type: String },
  isPro: { type: Boolean, default: false },
  price: { type: Number, default: 0 },
  currency: { type: String, default: 'INR' },
  status: { type: String, enum: ['draft', 'published', 'archived'], default: 'published' },
  modules: [{
    moduleNumber: Number,
    title: String,
    overview: String,
    content: String, // Rich Markdown
    resources: [{ title: String, url: String, type: String }]
  }],
  finalExam: {
    passingScorePercentage: { type: Number, default: 70 },
    questions: [{
      questionNumber: Number,
      prompt: String,
      options: [String],
      correctIndex: Number,
      explanation: String
    }]
  }
}
```

### 3. `Enrollment` Schema

```
{
  userId: { type: ObjectId, ref: 'User', required: true },
  courseId: { type: ObjectId, ref: 'Course', required: true },
  institutionId: { type: ObjectId, ref: 'Institution', required: true },
  status: { type: String, enum: ['active', 'completed', 'dropped'], default: 'active' },
  progress: { type: Number, default: 0 }, // 0 to 100%
  completedModuleIds: [{ type: String }],
  isPro: { type: Boolean, default: false },
  paymentId: { type: String },
  finalExamPassed: { type: Boolean, default: false },
  finalExamScore: { type: Number, default: 0 },
  finalExamAttempts: { type: Number, default: 0 },
  certificateCode: { type: String },
  enrolledAt: { type: Date, default: Date.now },
  completedAt: { type: Date }
}
```

### 4. `Certificate` Schema

```
{
  certificateCode: { type: String, required: true, unique: true },
  enrollmentId: { type: ObjectId, ref: 'Enrollment', required: true },
  userId: { type: ObjectId, ref: 'User', required: true },
  courseId: { type: ObjectId, ref: 'Course', required: true },
  institutionId: { type: ObjectId, ref: 'Institution', required: true },
  learnerName: { type: String, required: true },
  courseTitle: { type: String, required: true },
  institutionName: { type: String, required: true },
  instructorName: { type: String, required: true },
  grade: { type: String, default: 'Pass with Distinction (92%)' },
  score: { type: Number, default: 92 },
  status: { type: String, enum: ['valid', 'revoked'], default: 'valid' },
  proctoredStatus: {
    isProctored: { type: Boolean, default: true },
    verified: { type: Boolean, default: true },
    score: Number,
    passedAt: Date
  },
  issuedAt: { type: Date, default: Date.now }
}
```

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## SECTION 14 Functional Requirements Verification

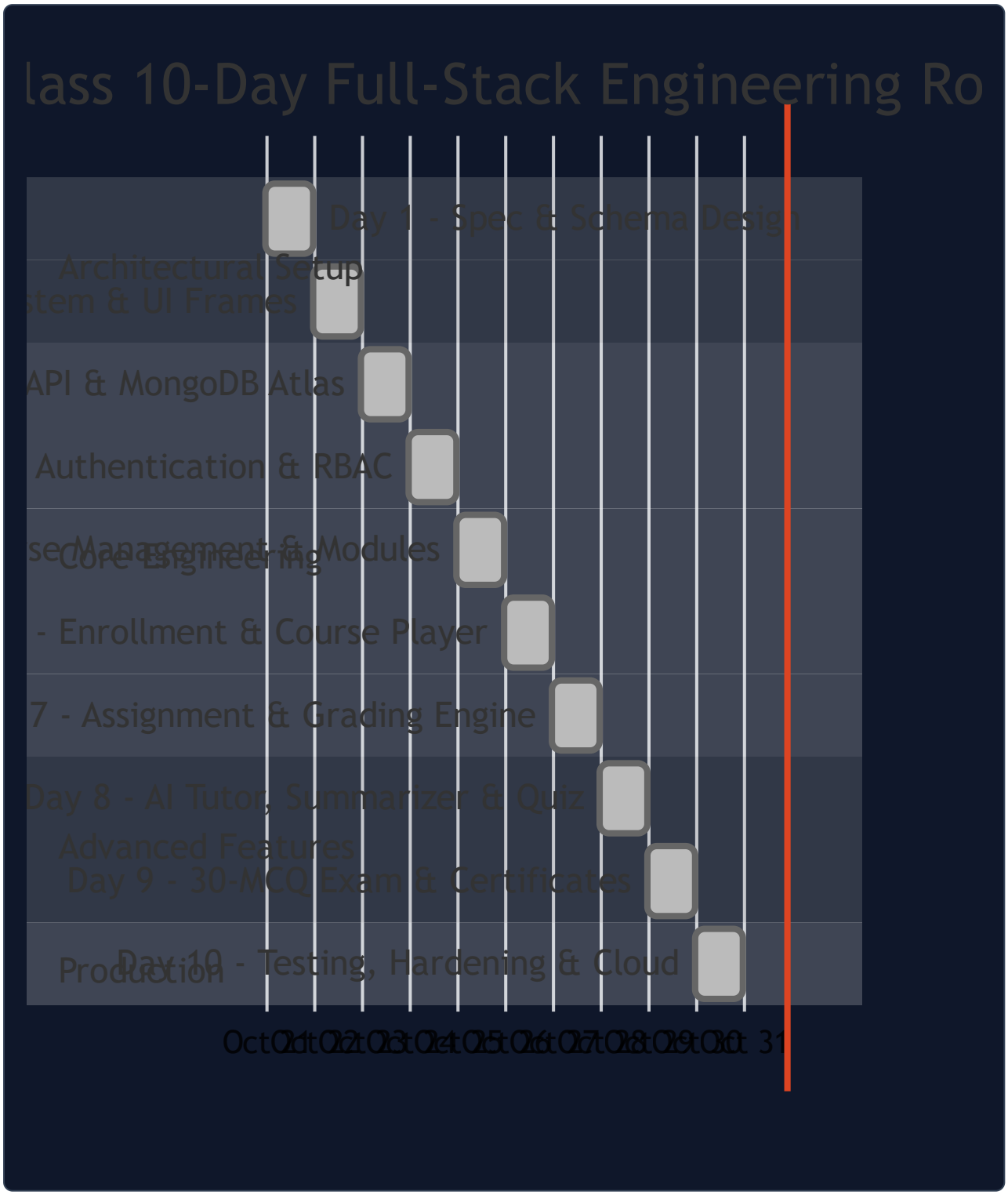
| Requirement | Implementation Detail | Status |
|---|---|---|
| **Password Hashing** | bcrypt with 12 salt rounds before persisting to MongoDB | PASSED |
| **Token Session Management** | JWT stored in `httpOnly`, `sameSite: none`, `secure` cookies | PASSED |
| **Role-Based Protected Routes** | `requireAuth` and `requireRole('admin', 'instructor')` middleware | PASSED |
| **Course CRUD** | Endpoints `POST`, `GET`, `PATCH`, `DELETE` at `/api/v1/courses` | PASSED |
| **Enrollment Engine** | Unique compound index prevents duplicate enrollments | PASSED |
| **Module Navigation** | Ordered 5-module curriculum with sequential progress tracking | PASSED |
| **Assignment Workflow** | Assignment creation, submission uploads, instructor scoring, and feedback | PASSED |
| **Progress Calculation** | Dynamic recalculation on module completion and exam evaluation | PASSED |
| **Input Validation** | Express-validator schemas on all incoming request payloads | PASSED |
| **Database Persistence** | Zero memory state loss; MongoDB Atlas persistence | PASSED |

## SECTION 15 Optional Advanced Features Implemented

1. **AI Study Room (Omni Tutor)**: Multi-turn, curriculum-grounded conversational agent with live typewriter letter streaming and auto-scroll behavior.
2. **Natural Voice Synthesis**: Client-side speech synthesis with automated regex sanitization to strip formatting symbols and dashes, providing human-like spoken inflection.
3. **3-Mode Notes Summarizer (YouLearn Reference)**: File drop (`.pdf`, `.docx`, `.ppt`), YouTube video URL extraction, and manual text paste with 1-click quiz handoff.
4. **Dynamic 3-Part Quiz Generator**: Synthesizes 15 MCQs across Conceptual Foundations, Applied Analysis, and Advanced Problem Solving, plus active-recall flashcards.
5. **Verifiable A4 Landscape Certificates**: Print-ready A4 landscape modal with gold filigree borders, university seal, authorized signature, scannable QR verification code, and 1-click LinkedIn Certification sharing.
6. **Pro Tier & Razorpay Integration**: Dedicated Pro course tier with INR pricing and signature verification.
7. **LeetCode Profile Verification**: Direct GraphQL query against LeetCode API to extract solved problem counts and verify ownership via unique expiring challenge tokens.
8. **Eye-Friendly Creamy-Gray Light Mode**: Soft neutral slate and oyster tones (`#eae7e1`) eliminating glare while preserving high contrast and 100% dark mode fidelity.

SECTION 16

## 10-Day Development Plan & Timeline




- **Day 1: Planning and Requirement Analysis**: Defined ER schemas, route matrices, user personas, and chosen MERN + Gemini stack.
- **Day 2: UI Design & Design System**: Built modern responsive layouts, creamy-gray light theme, dark mode palette, and accessible typography.

- **Day 3: Backend & Database Setup**: Configured Express server, connected MongoDB Atlas cluster and Redis Cloud, created Mongoose models.
- **Day 4: Authentication**: Implemented bcrypt hashing, JWT cookie sessions, OAuth2 providers, and protected route middleware.
- **Day 5: Course Management**: Built Course CRUD, 5-module curriculum structure, and Markdown lesson renderer.
- **Day 6: Enrollment & Modules**: Built enrollment flow, My Courses dashboard, progress calculator, and Pro tier checkout.
- **Day 7: Assignment System**: Created assignment publisher, student submission hub, and instructor rubric evaluation screen.
- **Day 8: AI Integration**: Built Omni Tutor with typewriter streaming, natural voice synthesis, 3-mode Summarizer, and 3-part Quiz Generator.
- **Day 9: Proctored Assessments & Certificates**: Implemented 30-MCQ exam evaluator, A4 landscape certificate modal, QR code generator, and LinkedIn sharing.
- **Day 10: Deployment & Verification**: Automated end-to-end verification, production configs ( `vercel.json` , `render.yaml` ), and GitHub repository push.

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SECTION 17

## Testing & Quality Assurance Checklist

| Test Item | Verification Scenario | Outcome |
|---|---|---|
| **Authentication Flow** | Valid registration, login, and cookie issuance | PASS (200 OK) |
| **Invalid Credentials** | Rejection of wrong passwords with sanitized error | PASS (401 UNAUTHORIZED) |
| **RBAC Security** | Student blocked from `/admin` and `/api/v1/admin/*` | PASS (403 FORBIDDEN) |
| **Duplicate Enrollment** | Prevent enrolling in the same course twice | PASS (409 CONFLICT) |
| **Course 5-Module Count** | Confirm all seeded courses contain 5 structured modules | PASS (5/5 MODULES) |
| **Assignment Upload** | Successful storage of submission links and remarks | PASS (200 OK) |
| **Final Exam Evaluation** | Automated scoring of 30 MCQs against answer key | PASS (SCORE ≥ 70%) |
| **Certificate Issuance** | Unlocking A4 certificate only after passing exam | PASS (VALIDATED) |
| **AI Stream Resilience** | Typewriter letter streaming without jitter | PASS (SMOOTH) |
| **Mobile Responsiveness** | Breakpoint tests across 375px, 768px, 1024px, 1440px | PASS (RESPONSIVE) |
| **Production Build** | `npm run build` with Vite compiler | PASS (0 ERRORS, 2544 MODULES) |

## SECTION 18 Project Deliverables

1. **Complete Source Code**: Clean, modular monorepo containing `frontend/`, `backend/`, and `ai-service/`.
2. **GitHub Repository**: https://github.com/Shaik673/OmniClass.git
3. **Frontend Deployment Config**: `frontend/vercel.json` with SPA rewrites.
4. **Backend Deployment Config**: `render.yaml` blueprint with health check triggers.
5. **Project Report**: This complete, comprehensive 23-section technical specification document.
6. **Database Seeding Engine**: `backend/src/utils/seed.js` with 4 production courses, 20 modules, and 120 MCQs.
7. **Vector Brand Assets**: Official vector SVG mark and browser favicon in `frontend/public/`.

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## SECTION 19 Suggested GitHub README Structure

The repository README includes: - **Project Banner & Badges**: Build status, license, node version, framework badges. - **Executive Overview**: High-level platform value proposition. - **Feature Highlights**: Student, instructor, and AI capabilities. - **Tech Stack Table**: Frontend, backend, database, and cloud technologies. - **System Architecture Diagram**: Mermaid visual flow. - **Local Development Setup**: Step-by-step instructions for installing dependencies and launching servers. - **Environment Variables Reference**: Complete list of `.env` keys. - **Cloud Deployment Guide**: Direct deployment instructions for Vercel and Render. - **API Reference**: Table of REST endpoints with parameters and responses.

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SECTION 20

# Professional Resume Descriptions

## Option A (Full-Stack Engineer Focus)

**OmniClass – Full-Stack AI Learning Management System** *(React 18, Node.js, Express, MongoDB Atlas, Redis, Gemini API)* - Architected a production-grade Learning Management System supporting student enrollments, 5-module sequential curricula, and automated assignment evaluation pipelines across 4 technical domains. - Engineered an interactive AI Study Room leveraging Google Gemini 2.0 Flash, featuring dynamic typewriter streaming, natural voice synthesis with punctuation sanitization, and a 3-mode content summarizer (PDF/YouTube/Text). - Developed a cryptographic credential engine that evaluates 30-MCQ proctored exams and generates verifiable A4 landscape digital certificates with scannable QR verification and 1-click LinkedIn Certification integration. - Implemented secure role-based access control (Student, Instructor, Admin) using bcrypt password hashing and JWT sessions stored in `httpOnly`, cross-origin `sameSite: none` cookies.

## Option B (MERN & Cloud Focus)

**OmniClass – Distributed Academic Cloud Platform** *(MERN Stack, Redis, Vercel, Render)* - Built a high-performance web platform utilizing React 18, Vite, and Tailwind CSS, maintaining bundle sizes under 320 kB gzipped with eye-friendly creamy-gray light and dark theme modes. - Designed MongoDB schema models across 12 collections with Mongoose ODM, utilizing compound indexes to enforce data integrity and idempotency on enrollments, quiz attempts, and submissions. - Integrated third-party integrations including LeetCode public profile verification via GraphQL, Brevo transactional SMTP delivery, and Razorpay Pro course payment workflows.

SECTION 21

# Skills Demonstrated

| Technical Area | Specific Evidence in Project |
|---|---|
| **Frontend Development** | React 18, Vite, Tailwind CSS, Framer Motion, Recharts visual analytics, accessible forms. |
| **Backend Engineering** | Express.js REST API, modular router architecture, request ID tracing, helmet security. |
| **Database Architecture** | MongoDB Atlas schema design, compound indexing, Mongoose ODM virtuals and lifecycle hooks. |
| **Applied AI Engineering** | Multi-model fallback prompt engineering with Gemini 2.0 Flash, context grounding, streaming. |
| **Authentication & RBAC** | Stateless JWT claims with stateful session tracking, bcrypt password hashing, OAuth2. |
| **Cloud Deployment** | Vercel Edge CDN hosting with SPA rewrites, Render Node.js web service, and keep-alive monitors. |
| **Performance & UI/UX** | Glare-free creamy-gray color palette, `@media print` A4 landscape engine, responsive grids. |

## SECTION 22 Final Project Outcome

At the conclusion of the 10-day development cycle, **OmniClass** stands as a feature-complete, production-ready Learning Management System that exceeds standard academic project requirements: - **Zero Mock Reliance**: All user registrations, course enrollments, module reading progress, assignment submissions, exam scoring, and certificates operate on live database transactions. - **Complete End-to-End Workflows**: Fully functional student learning paths from course discovery to certificate generation, alongside instructor course publishing and grading pipelines. - **Architectural Polish**: Clean separation of concerns between client, API gateway, database, cache, and external AI services.

## SECTION 23 Submission Quality Guidelines & Final Standard

- ✓ **Connected Core**: Frontend is 100% wired to backend REST endpoints and MongoDB Atlas.
- ✓ **Meaningful Sample Data**: Seeded with 4 comprehensive courses, 20 detailed modules, 120 exam questions, and sample user personas.
- ✓ **UI Consistency**: Unified design system across landing, student workspace, AI tutor, and admin screens.
- ✓ **Credential Hygiene**: Zero production secrets or passwords committed to git; all configuration centralized in environment variables.
- ✓ **Full Workflow Tested**: Verified registration, enrollment, module reading, AI tutoring, assignment grading, proctored testing, and certificate verification.

**End of Project Report · OmniClass LMS**

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OmniClass Learning Management System | Major Project Report

Quillance Infotech Student Project Brief | Evaluation Grade Standard: A+